

A finite-volume implementation of the phase-field model for brittle fracture with adaptive mesh refinement

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ABSTRACT

The phase-field method (PFM) has advantages in modeling crack propagation in rock-like materials by treating the crack surface as a continuous function, therefore avoiding dealing with the sharp discontinuous displacement field. This study presents an implementation of PFM using the finite volume method (FVM) to discretize the governing equations for the conservation of linear momentum and the evolution of phase field. The coupled equations are solved by an iterative staggered scheme. The adaptive mesh refinement (AMR) technique is further employed to improve computational efficiency, which is relatively easy to achieve in FVM as the method can naturally handle unstructured meshes with hanging nodes in both two- and three-dimensional problems, as opposed to the finite element method (FEM). Several classical crack propagation problems, including the single-edge notched tension and shear tests, the L-shaped panel test, and the test on a notched plate with a hole, are simulated and compared with available experimental and FEM results, which demonstrate that the FVM-based PFM can model crack propagation accurately and effectively and could be more efficient than the FEM-based PFM.

1. Introduction

Modeling crack propagation in rock-like materials has attracted widespread interest but remains challenging due to the material heterogeneity involving inherent defects at different scales and directions, as well as the complexity of the problem itself. Being an essentially discontinuous process, the crack propagation problem poses tremendous difficulties for the traditional continuum-based numerical methods, especially the mesh-based methods such as the finite element method (FEM) that is still dominant in computational solid mechanics.

Presently, two conceptually different methods are mainly used in FEM to tackle crack propagation, corresponding to the discontinuous (discrete) and the continuous (smeared) categories, respectively. The former describes the crack through the discontinuous displacement field, for example, by simply deleting the elements that satisfy the crack condition (Belytschko and Lin, 1987). This element deletion method (EDM) however cannot simulate the branch of crack correctly (Liu et al., 2014). Another option is to insert cohesive interface elements at the element interfaces, so the crack can propagate through the interfaces

(Xu and Needleman, 1994). Since crack propagation inside the elements is not permitted, this method has strong mesh dependence. On the other hand, the extended finite element method (XFEM) (Belytschko and Black, 1999) has proven to be one of the most well-received discrete methods so far, which introduces an enriched basis function to describe the displacement discontinuity thus allowing the crack to propagate within an element and can track the crack path without remeshing. Nevertheless, the tracking of crack surface is a formidable task, especially in 3D problems.

In contrast to the discrete methods that explicitly simulate the strong discontinuity in displacements across cracks, the continuous methods model the crack as a diffuse zone, so that the displacement fields remain continuous, and the tracking of crack surface is no longer required. An example of such methods is the continuum damage mechanics (CDM) model (Bazant and Cedolin, 1983), which expresses the progressive damage of a material through the evolution of some internal variables such as the damage variable. Yet the evolution law of the damage variable is often phenomenological and simply assumed to be a function of the equivalent plastic strain. Therefore, it is difficult for CDM to

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accurately predict the crack initiation and propagation path. More recently, the phase field method (PFM) (Aranson et al., 2000) has emerged as a competitive alternative for modeling crack propagation and can deal with single and multiple cracks with complex topologies in both 2D and 3D conditions.

The PFM bears sound physical background rooted in the variational approach proposed by Francfort and Marigo (1998) based on the Griffith theory of fracture (Griffith, 1921). In the regularized variational formulation (Bourdin et al., 2000; Tanné et al., 2018), the sharp crack surface Γ is regularized as a band with a certain length scale and described by an order parameter, i.e., the phase-field variable. Previous studies (e.g., Miehe et al., 2010a) have shown that when the length scale tends to 0, the diffuse crack converges to a sharp crack. In the past decade, many versions of PFM have been developed. For instance, Amor et al. (2009) and Miehe et al. (2010a) successively proposed the spherical-deviatoric decomposition of strain energy and the spectral decomposition of strain energy to distinguish the effects of tensile and compressive stresses. Borden et al. (2014) proposed a fourth-order phase-field model that leads to higher regularity in the exact phase-field solution and a faster convergence rate. Ambati et al. (2015) proposed a hybrid isotropic-anisotropic phase-field formulation achieving a significant reduction of computational cost. Furthermore, many phase-field models tailored for specific materials have been developed to investigate crack propagation in various scenarios. Zhang et al. (2017) proposed a modified phase-field model using different critical energy release rates for different fracture modes and applied it to the study of mixed-mode crack in rock-like materials. In addition, Fei and Choo (2019) creatively introduced a stress decomposition method based on the interface stress to model rock fracture propagation considering frictional contact. They have applied this model in geotechnical engineering problems such as the slope stability analysis (Fei and Choo, 2020). Li et al. (2023b) introduced an adaptive phase-field method based on the isogeometric-meshfree approach to capture the dynamic crack propagation in composite materials under impact loading. Kiran et al. (2022, 2023) developed an adaptive phase-field model for the study of electromechanical fracture of piezoelectric ceramics and the interfacial fracture in transversely isotropic piezoelectric materials under various electromechanical loading conditions. For the solution of the coupled displacement and phase field governing equations, the staggered scheme (Seles et al., 2019) is usually adopted due to its proved robustness.

Most of the existing phase-field models have been implemented in FEM, whereas a recent study by Sargado et al. (2021) has demonstrated a notable gain in computational efficiency by using a combined FEM-finite volume method (FVM), where the stress equilibrium equation is discretized by standard FEM and the phase-field evolution equation by FVM. Their study has directly motivated the present study to implement the phase-field model entirely in FVM considering that FVM is equally applicable to solve linear and nonlinear solid mechanics problems (Demirdžić et al., 1998; Demirdžić and Martinović, 1993; Demirdžić and Muzaferija, 1994; Demirdžić and Muzaferija, 1995; Demirdžić et al., 1997). Cardiff et al. (2018) developed an open-source FVM toolbox called solids4foam for solid mechanics and fluid-structure interaction problems based on OpenFOAM (Weller et al., 1998), which will be used in the present study. The unified FVM implementation of both stress equilibrium and phase field evolution equations will lead to several other benefits. For example, the cell-centered FVM used in the study finds solutions from conservation equations on local cells, thus allowing for jumps in solutions at the cell interfaces. Consequently, it can conveniently handle general structured or unstructured polyhedral meshes with hanging nodes, which means that the method is friendly to adaptive mesh refinement (AMR) (Meredith and Vukčević, 2018), a desired feature for phase field modeling, especially when the crack path is not known *a priori*, thus predetermined local mesh refinement (Wu et al., 2021) is infeasible. In the literature, the phase field (Borden et al., 2012) or its gradient (Li et al., 2023a) is commonly used as the

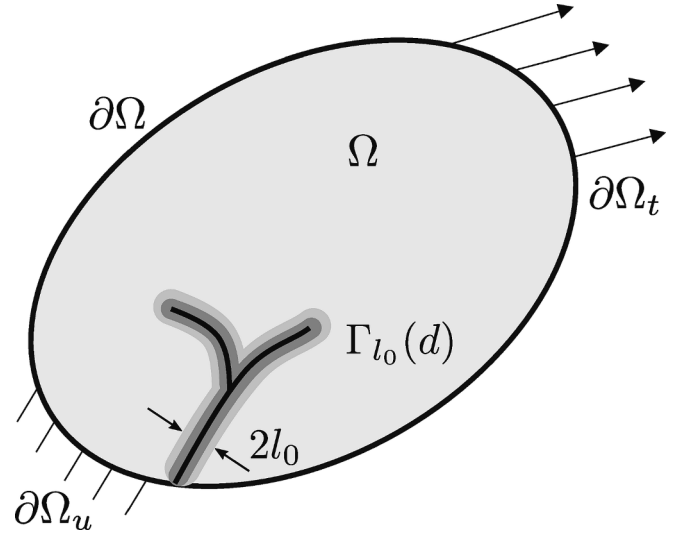


Fig. 1. Schematic diagram of diffuse cracks.

refinement index. In this regard, the FVM is more flexible than the FEM in terms of mesh generation and adaptive remeshing. Furthermore, the FVM is naturally conservative and particularly suitable for solving continuity or transport equations, therefore popular in computational fluid dynamics. As a result, the proposed FVM-based phase-field model could be potentially conveniently coupled with fluid flow to solve multiphysics problems such as hydraulic fracturing.

The rest of the paper is organized as follows. Section 2 provides an overview of the phase field formulation for brittle fracture with the two strain energy decomposition methods proposed by Amor et al. (2009) and Miehe et al. (2010a), respectively. Section 3 presents the FVM discretization of the governing equations for the balance of linear momentum and phase field evolution. Next, seven numerical examples including the single-edge notched tension and shear tests, the L-shaped panel test, and the test on a notched plate with a hole, are given in Section 4 to demonstrate the correctness and efficiency of the proposed model. Section 5 concludes the study.

2. Phase field formulation

In this section, we briefly outline a typical phase field formulation for brittle fracture, which is mainly based on the variational formulation in Miehe et al. (2010a) and derived from the classical fracture theory (Griffith, 1921) and the minimum energy variational method (Francfort and Marigo, 1998). The former assumes that the crack propagates only when the energy release rate reaches a critical value, whereas the latter assumes that the crack propagation process should minimize the free energy and follow irreversibility conditions. In the PFM, the crack is described by a diffuse phase boundary, which is represented by a continuous function of the order parameter d . The value of d is in the range $[0, 1]$, with $d = 0$ for the intact region and $d = 1$ for the fully damaged region.

Consider a brittle linear elastic solid $\Omega \subset \mathbb{R}^n$ (n denotes the number of dimensions) with an embedded fracture Γ as shown in Fig. 1, whose outer boundary is $\partial\Omega \subset \mathbb{R}^{n-1}$ and subjected to either the Dirichlet boundary condition $\partial\Omega_u$ or the Neumann boundary condition $\partial\Omega_t$, satisfying $\partial\Omega_u \cup \partial\Omega_t = \partial\Omega$ and $\partial\Omega_u \cap \partial\Omega_t = \emptyset$ (applies to each component for the vector fields). The crack fracturing processes can be determined using the following potential energy functional:

$$\Pi(\mathbf{u}, \Gamma) = \int_{\Omega} \Psi(\varepsilon(\mathbf{u})) d\Omega + \int_{\Gamma} G_c d\Gamma \quad (1)$$

where Ψ is the elastic strain energy density, $\varepsilon(\mathbf{u}) = ((\nabla \mathbf{u})^T + \nabla \mathbf{u})/2$ is

the small strain tensor, G_c is the critical energy release rate of the material, and $\mathbf{u}$ is the displacement. The strain energy of an isotropic linear elastic material can be defined by the following formula:

$$\Psi(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} \text{tr}(\boldsymbol{\varepsilon})^2 + \mu \boldsymbol{\varepsilon} : \boldsymbol{\varepsilon} \quad (2)$$

where λ and μ are the Lamé constants. With the phase field variable d , the surface density of cracks per unit volume γ can be expressed as (Miehe et al., 2010b):

$$\gamma(d) = \frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \quad (3)$$

where ∇d is the spatial gradient of the phase field, and $l_0 \in R^+$ is the characteristic length that controls the width of the diffuse crack transition region, as shown in Fig. 1. Therefore, the regularized crack surface $\Gamma_{l_0}(d)$ can be represented by a scalar phase field via the following integral:

$$\Gamma_{l_0}(d) = \int_{\Omega} \gamma(d) d\Omega \quad (4)$$

The total crack surface energy of the elastic solid can be obtained from Eq. (4):

$$\int_{\Gamma} G_c d\Gamma \approx \int_{\Omega} G_c \left(\frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right) d\Omega \quad (5)$$

Therefore, the expansion of the total potential energy can be obtained as

$$\Pi(\mathbf{u}, d, \nabla d) = \int_{\Omega} g(d) \Psi(\boldsymbol{\varepsilon}(\mathbf{u})) d\Omega + \int_{\Omega} G_c \left(\frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right) d\Omega \quad (6)$$

where $g(d)$ is the stress degradation function that reduces the stored energy within the solid as the phase field d evolves. Here, it is specified as $g(d) = [(1-d)^2 + k]$, where k is a model parameter to avoid numerical singularity when d approaches unity and taken as $k = 10^{-6}$ throughout the study. Its influence is proved negligible according to the existing studies, e.g., Miehe et al. (2010a).

The external force work W_{ext} is expressed as (body force neglected in the study):

$$W_{\text{ext}} = \int_{\partial\Omega_t} \mathbf{t} \cdot \mathbf{u} dA \quad (7)$$

where $\mathbf{t}$ is the surface traction. According to Eqs. (6) and Eq. (7), the governing equations of the phase field model can be obtained by minimizing the total energy functional, i.e., letting the variation of Eq. (8) be zero.

$$\Pi - W_{\text{ext}} = \int_{\Omega} g(d) \Psi(\boldsymbol{\varepsilon}(\mathbf{u})) d\Omega + \int_{\Omega} G_c \left(\frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right) d\Omega - \int_{\partial\Omega_t} \mathbf{t} \cdot \mathbf{u} dA \quad (8)$$

As a result, the following coupled partial differential equations are reached:

$$\nabla \cdot \boldsymbol{\sigma} = 0 \text{ in } \Omega \quad (9a)$$

$$-G_c l_0 \nabla^2 d + \left[\frac{G_c}{l_0} + 2\Psi(\boldsymbol{\varepsilon}) \right] d = 2\Psi(\boldsymbol{\varepsilon}) \text{ in } \Omega \quad (9b)$$

with the following boundary conditions:

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} \text{ on } \partial\Omega_t \quad (10a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \text{ on } \partial\Omega_u \quad (10b)$$

$$\nabla d \cdot \mathbf{n} = 0 \text{ on } \partial\Omega \quad (10c)$$

where $\bar{\mathbf{t}}$ and $\bar{\mathbf{u}}$ are the prescribed boundary traction and displacement, respectively. $\mathbf{n}$ is the outward normal vector. The stress is

calculated as:

$$\boldsymbol{\sigma} = [(1-d)^2 + k] \frac{\partial \Psi(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}} \quad (11)$$

Existing studies have shown that the nucleation and propagation of cracks are often caused by tension, whereas compression will lead to the closure of cracks and hinder their propagation. Hence, some tension-compression decomposition schemes are developed. A comparison of different decomposition methods can be found in Ambati et al. (2015). In general, they can be classified as isotropic and anisotropic decompositions. The isotropic ones (Bourdin et al., 2000; Yang, 2006) are linear and therefore less computationally expensive, but they allow cracks to grow in compression and may lead to nonphysical crack patterns. On the contrary, the anisotropic formulations decompose the strain energy density into positive and negative parts, and the stress equation is reformulated as:

$$\boldsymbol{\sigma} = [(1-d)^2 + k] \frac{\partial \Psi^+(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}} + \frac{\partial \Psi^-(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}} \quad (12)$$

The commonly used decomposition methods are the spherical-deviatoric decomposition proposed by Amor et al. (2009) and the spectral decomposition proposed by Miehe et al. (2010a). For the spherical-deviatoric decomposition, the strain energy can be expressed as:

$$\Psi^+ = \frac{1}{2} \left(\lambda + \frac{2\mu}{n} \right) \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_+^2 + \mu (\boldsymbol{\varepsilon}_{\text{dev}} : \boldsymbol{\varepsilon}_{\text{dev}}) \quad (13a)$$

$$\Psi^- = \frac{1}{2} \left(\lambda + \frac{2\mu}{n} \right) \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_-^2 \quad (13b)$$

where $\boldsymbol{\varepsilon}_{\text{dev}}$ is the deviatoric part of the strain tensor. The Macaulay brackets are defined as: $\langle x \rangle_{\pm} = (x \pm |x|)/2$. For the spectral decomposition method, the strain tensor is decomposed into positive and negative parts:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_+ + \boldsymbol{\varepsilon}_- \quad (14)$$

$$\boldsymbol{\varepsilon}_+ = \sum_{p=1}^n \langle \varepsilon_p \rangle_+ \mathbf{n}_p \otimes \mathbf{n}_p \quad (15a)$$

$$\boldsymbol{\varepsilon}_- = \sum_{p=1}^n \langle \varepsilon_p \rangle_- \mathbf{n}_p \otimes \mathbf{n}_p \quad (15b)$$

where $\{\varepsilon_p\}_{p=1 \dots n}$ and $\{\mathbf{n}_p\}_{p=1 \dots n}$ are the principal strains and their directions, respectively. Therefore, the positive and negative parts of elastic energy can be expressed as:

$$\Psi^+ = \frac{\lambda}{2} \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_+^2 + \mu \text{tr}[\boldsymbol{\varepsilon}_+^2] \quad (16a)$$

$$\Psi^- = \frac{\lambda}{2} \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_-^2 + \mu \text{tr}[\boldsymbol{\varepsilon}_-^2] \quad (16b)$$

The elastic energy can be expressed by the following formula:

$$\Psi = g(d) \Psi^+ + \Psi^- \quad (17)$$

Substituting Eq. (17) into Eq. (8), and taking the variation with respect to $\mathbf{u}$ and d , the governing equations can be obtained:

$$\nabla \cdot \boldsymbol{\sigma} = 0 \text{ in } \Omega \quad (18a)$$

$$-G_c l_0 \nabla^2 d + \left[\frac{G_c}{l_0} + 2\Psi^+ \right] d = 2\Psi^+ \text{ in } \Omega \quad (18b)$$

where the stress $\boldsymbol{\sigma}$ is given by Eq. (12).

It is necessary to supplement the irreversibility condition to prevent the crack healing of the elastic solid under compression or unloading. The commonly used method is to introduce a history state variable H , which is updated according to:

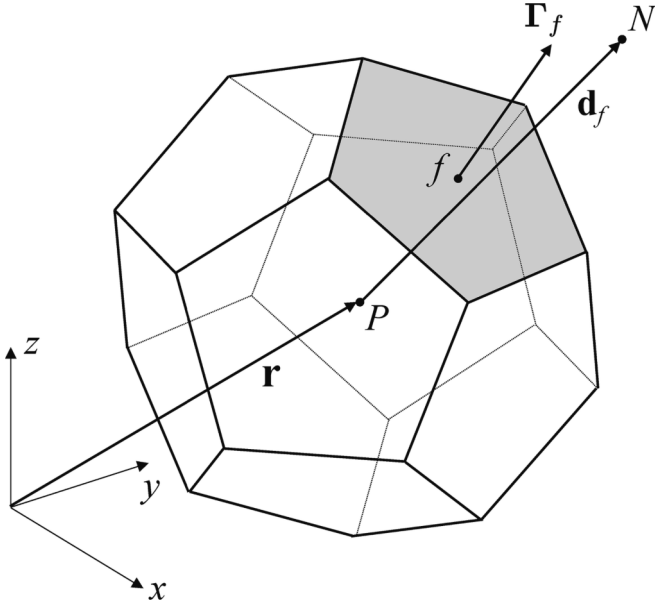


Fig. 2. Polyhedral control volume in FVM (after Cardiff et al., 2014).

$$H_{n+1} = \max(H_n, \Psi_{n+1}^+) \quad (19)$$

It represents the maximum tensile elastic strain energy from the start of loading to the current time t_{n+1} . Replacing the elastic energy Ψ^+ in Eq. (18b) with H leads to a non-decreasing d . Then, Eq. (18b) is rewritten as:

$$-G_c l_0 \nabla^2 d + \left[\frac{G_c}{l_0} + 2H \right] d = 2H \text{ in } \Omega \quad (20)$$

In summary, Eqs. (18a), (19) and (20) complete the phase field model. It is worth noting that the two different decomposition methods use either Eq. (13a) or (16a) to calculate Ψ^+ . The spectral decomposition is mainly used in this study unless otherwise stated.

3. Finite volume implementation

We use the residual control staggered solution scheme (Seleš et al., 2019) to alternately solve the displacement and phase field coupling equations by the implicit cell-centered FVM (Cardiff and Demirdžić, 2021). FVM converts a given partial differential equation representing the conservation law into a system of discrete algebraic equations on a finite control volume (CV). Like in other mesh-based methods, the geometric domain is first discretized into non-overlapping cells or finite volumes. Next, the transformed algebraic equation system is solved to find the field variables of each cell. The discretization of the solution domain and equations in a standard cell-centered FVM and the solution procedure are introduced below.

3.1. Discretization

The computational space is divided into a finite number of convex polyhedral CVs surrounded by convex polygon faces. The discretization of variables in space is generally linear, so the FVM discretization is usually second-order accurate (Cardiff et al., 2018). A typical CV is shown in Fig. 2, where the point P is located at the center of the CV with the position vector $\mathbf{r}$. N is the center of the neighboring CV that shares the face f with P . Γ_f is the area normal vector of f , i.e., $|\Gamma_f|$ is the face area, and $\mathbf{d}_f$ is the vector connecting P and N .

The mesh is considered non-orthogonal if Γ_f and $\mathbf{d}_f$ are not parallel and skewed if $\mathbf{d}_f$ does not intersect f at its center. The non-orthogonal or skewed mesh will deteriorate the accuracy of FVM. Remedies are often required to optimize the discretization, which are detailed in Jasak

(1996).

By integrating Eq. (18a) over the volume of P , Ω_P , and applying the divergence theorem, it gives:

$$\int_{\Omega_P} \nabla \cdot \boldsymbol{\sigma} d\Omega = \oint_{\partial\Omega_P} \mathbf{n} \cdot \boldsymbol{\sigma} dA = 0 \quad (21)$$

Given the stress term on the face σ_f , the surface integral in Eq. (21) can be approximated by:

$$\oint_{\partial\Omega_P} \mathbf{n} \cdot \boldsymbol{\sigma} dA \approx \sum_f \Gamma_f \cdot \boldsymbol{\sigma}_f \quad (22)$$

where the summation is over the faces of cell P .

Taking the spectral decomposition method for example, σ_f can be obtained from Eqs. (12) and (16) as:

$$\boldsymbol{\sigma}_f = g(d) \{ \lambda \langle \text{tr}(\boldsymbol{\epsilon}_f) \rangle_+ \mathbf{I} + 2\mu \langle \boldsymbol{\epsilon}_f \rangle_+ \} + \lambda \langle \text{tr}(\boldsymbol{\epsilon}_f) \rangle_- \mathbf{I} + 2\mu \langle \boldsymbol{\epsilon}_f \rangle_- \quad (23)$$

where the subscript f indicates the quantity evaluated at the face center, and $\mathbf{I}$ is the identity tensor. In the cell-centered FVM, the displacement is assumed to vary linearly within a cell:

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}_P + (\mathbf{x} - \mathbf{x}_P) \cdot (\nabla \mathbf{u})_P \quad (24)$$

where $\mathbf{u}_P$ and $(\nabla \mathbf{u})_P$ are the center displacement and constant displacement gradient of the cell, respectively. The constant displacement gradient (or strain) will result in a discontinuity at the face of neighboring cells. Hence, an interpolation weight method is used to evaluate the quantity at the face shared by cells P and N :

$$(\nabla \mathbf{u})_f \approx \gamma_f (\nabla \mathbf{u})_P + (1 - \gamma_f) (\nabla \mathbf{u})_N \quad (25)$$

where the weight γ_f is in the range (0, 1) and usually obtained by the following formula:

$$\gamma_f = \frac{|\mathbf{x}_N - \mathbf{x}_f|}{|\mathbf{x}_N - \mathbf{x}_P|} \quad (26)$$

where $\mathbf{x}_P$, $\mathbf{x}_f$, and $\mathbf{x}_N$ are the coordinates of P , f , and N , respectively, as shown in Fig. 2.

The displacement gradient at the cell center is usually obtained using the least squares method due to its ease of implementation on unstructured grids:

$$(\nabla \mathbf{u})_P \approx \left[\sum_f w_f^2 \mathbf{d}_f \mathbf{d}_f \right]^{-1} \sum_f \left[w_f^2 \mathbf{d}_f (\mathbf{u}_N - \mathbf{u}_P) \right] \quad (27)$$

where the weight w_f can be taken as either 1 (Demirdžić and Muzaferija, 1995) or $|\mathbf{d}_f|^{-1}$ (Jasak and Weller, 2000). To reach the implicit formulation, the separation solution algorithm is adopted (Jasak and Weller, 2000):

$$\underbrace{K_f \nabla^2 \mathbf{u}}_{\text{implicit}} - \underbrace{K_f \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\sigma}}_{\text{explicit}} = 0 \quad (28)$$

where the coefficient K_f controls the smoothing effect and is typically taken as $K_f = \lambda + 2\mu$ (Cardiff et al., 2017). The implicit term in Eq. (28) will be used to assemble the matrix (see $[A^u]$ in Eq. (36a)) of the resulting algebraic linear system, and the explicit terms contribute to the source vector (see $[b^u]$ in Eq. (36a)) of the linear equation system using known values from the last iterative step.

The discretization however may suffer from the so-called checkerboarding errors (Cardiff and Demirdžić, 2021), with displacement oscillations close to the spurious singular modes that appear in the reduced-integration FEM. A diffusion term based on the Rhie-Chow correction (Rhie and Chow, 1983) is usually added in the implicit cell-centered scheme for stabilization, shown as the second term in the left-hand side of Eq. (29). The updated discretized linear momentum equation becomes:

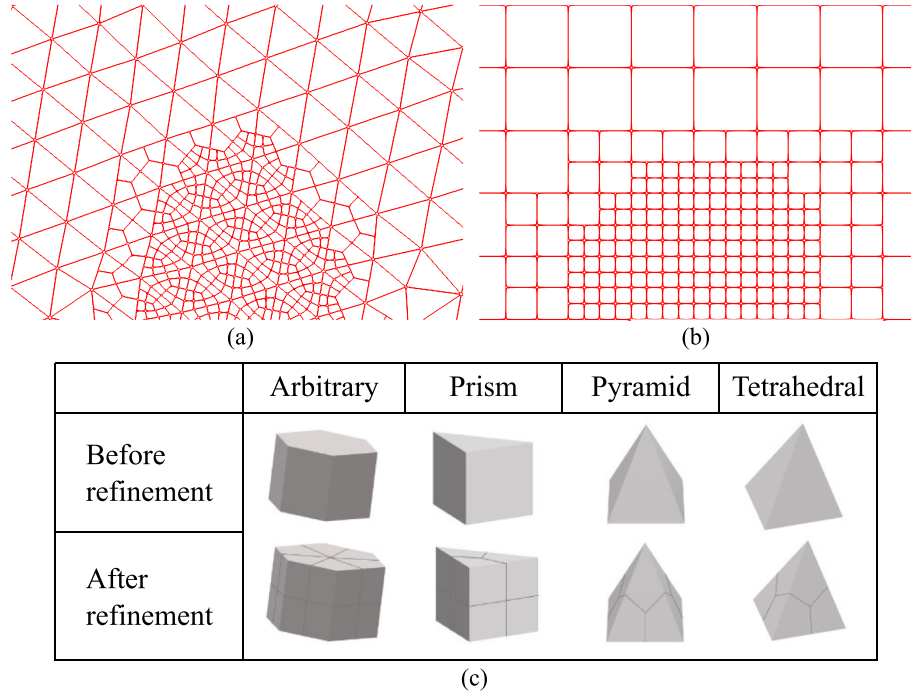


Fig. 3. Adaptive mesh refinement for (a) triangle and (b) quadrilateral cells in 2D, and (c) polyhedral cells in 3D (after Meredith and Vukčević, 2018).

$$\sum_f \Gamma_f \cdot \sigma_f + \underbrace{\sum_f (\lambda + 2\mu) \left[\frac{|\Delta_f|}{|\mathbf{d}_f|} \frac{\mathbf{u}_N - \mathbf{u}_P}{|\mathbf{d}_f|} - \Delta_f \cdot (\nabla \mathbf{u})_f \right]}_{\text{Rhie-Chow correction term}} = 0 \quad (29)$$

where

$$\Delta_f = \frac{\mathbf{d}_f}{\mathbf{d}_f \cdot \Gamma_f} |\Gamma_f|^2 \quad (30)$$

For the phase field equation, Eq. (20) is modified to:

$$\underbrace{-l_0^2 \nabla^2 d + \left(\frac{2Hl_0}{G_c} + 1 \right) d}_{\text{implicit}} = \frac{2Hl_0}{G_c} \quad (31)$$

By integrating Eq. (31) over Ω_P and applying the divergence theorem, it gives:

$$\int_{\Omega_P} \left(\frac{2Hl_0}{G_c} + 1 \right) d \, d\Omega + \int_{\partial\Omega_P} (-l_0^2 \nabla d \cdot \mathbf{n}) \, dA = \int_{\Omega_P} \frac{2Hl_0}{G_c} \, d\Omega \quad (32)$$

The discrete equation can be obtained as:

$$\left(\frac{2Hl_0}{G_c} + 1 \right) d_P \Omega_P - \sum_f l_0^2 (\nabla d)_f \cdot \Gamma_f = \frac{2Hl_0}{G_c} \Omega_P \quad (33)$$

where $(\nabla d)_f \cdot \Gamma_f$ can be approximated by:

$$(\nabla d)_f \cdot \Gamma_f = |\Gamma_f| \frac{d_N - d_P}{|\mathbf{d}_f|} \quad (34)$$

3.2. Boundary conditions and system of equations

The boundary conditions in FVM can be classified into three types according to Jasak and Weller (2000):

- For the *displacement* boundary, directly specify the displacement $\mathbf{u}$ on the boundary face. The surface gradient can be calculated from the center values of adjacent cells, e.g., Eq. (25).
- For the *symmetric* boundary, image the control volume on the other side of the boundary as an adjacent control volume.

- For the *traction* boundary, specify forces on the boundary surfaces:

$$\mathbf{g}_b = |\Gamma_b| \mathbf{t} \quad (35)$$

where Γ_b is the outward-pointing area vector of the boundary face, $\mathbf{t}$ is the specified traction. $\mathbf{g}_b$ will be added to Eq. (22) in calculating the force balance of the control volume.

After the above discretization, the discrete forms of Eqs. (18a) and (20) on each control volume can be cast into four (for 3D) linear algebraic equations (one vector equation for the displacement and one scalar equation for the phase field). The equation system for the whole domain can be obtained by assembling all the CVs, which presents in the following general form:

$$[A^u][\mathbf{u}] = [b^u] \quad (36a)$$

$$[A^d][d] = [b^d] \quad (36b)$$

where $[A^u]$ and $[A^d]$ are both assembled sparse matrices with diagonal terms a_P^u and a_P^d and non-zero neighbor off-diagonal terms a_N^u and a_N^d . $[b^u]$ and $[b^d]$ are the source vectors with terms b_P^u and b_P^d which cover the contribution from the explicit terms and boundary conditions. The expressions of these terms are shown below:

$$a_N^u = -(2\mu + \lambda) \frac{|\Gamma_f|}{|\mathbf{d}_f|} \quad (37a)$$

$$a_P^u = -\sum_f a_N^u \quad (37b)$$

$$a_N^d = -l_0^2 \frac{|\Gamma_f|}{|\mathbf{d}_f|} \quad (37c)$$

$$a_P^d = \left(\frac{2Hl_0}{G_c} + 1 \right) \Omega_P - \sum_f a_N^d \quad (37d)$$

$$b_P^u = \sum_f \Gamma_f \cdot \sigma_f - \sum_f (2\mu + \lambda) \Delta_f \cdot (\nabla \mathbf{u})_f \quad (37d)$$

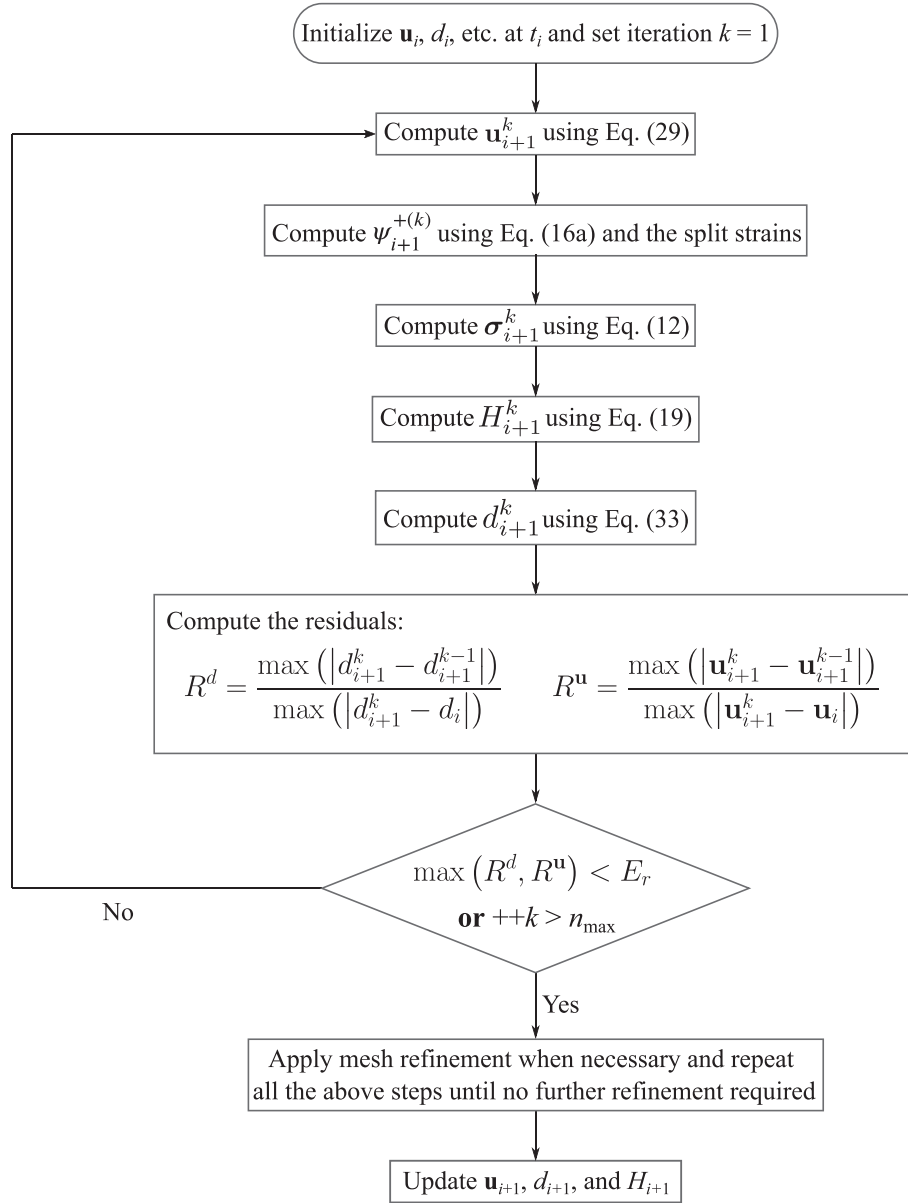


Fig. 4. Iterative staggered scheme of the FVM-based phase field model with adaptive mesh refinement.

$$b_p^d = \frac{2Hl_0}{G_c} \Omega_p \quad (37f)$$

Eq. (36) can then be assembled from each CV using the above equations as:

$$a_p^u \mathbf{u}_p + \sum_f a_N^u \mathbf{u}_N = b_p^u \quad (38a)$$

$$a_p^d d_p + \sum_f a_N^d d_N = b_p^d \quad (38b)$$

It is worth noting that the model implementation is based on solid4foam (Cardiff et al., 2018), an OpenFOAM toolbox for solid mechanics. The preconditioned conjugate gradient solver is used to solve Eq. (36) in the study with default settings.

3.3. Adaptive mesh refinement and solution flow

When a discrete crack is modeled as a diffuse phase field, abrupt gradient changes occur in the phase field. It is then necessary to use very

fine meshes to capture the abrupt gradient changes, often leading to a surge in computational cost especially when the crack path is unknown *a priori*. Therefore, the adaptive mesh refinement technique (Gupta et al., 2022; Krishnan et al., 2022; Xu et al., 2022; Dinachandra and Alankar, 2022; Patil et al., 2018; Assaf et al., 2022) is preferred. Compared with FEM, FVM can naturally deal with polyhedral or polygonal cells with hanging nodes, which means that it is more conducive to applying adaptive mesh refinement in FVM. The isotropic refinement algorithm proposed by Meredith and Vukčević (2018) for OpenFOAM has been used in this study. Generally, it first identifies the candidate cells that meet a certain refinement criterion ($d \geq 0.4$ in the study) and then splits the cells into a set of sub-cells by inserting points at the centers of the cells, the faces, and the edges. The process of point insertion will be repeated according to the maximum refinement level, and the sub-cells will be marked so that they will not be further refined in later loading steps. The refinement in 2D with initial triangle and quadrilateral cells after local refinement are shown in Fig. 3(a) and (b), respectively. Similarly, the refinement in 3D for different polyhedrons is illustrated in Fig. 3(c). Note that a posteriori error estimator (Walloth and Wollner,

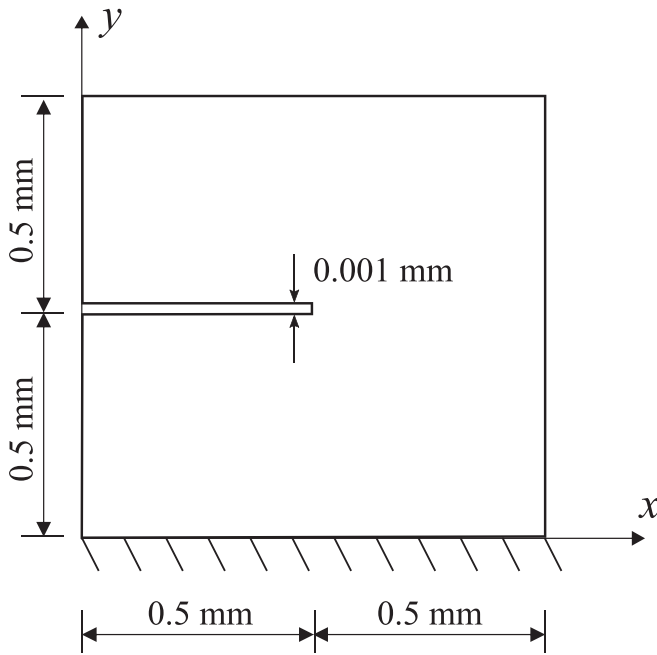


Fig. 5. Geometry and boundary conditions for the single-edge notched tension and shear tests.

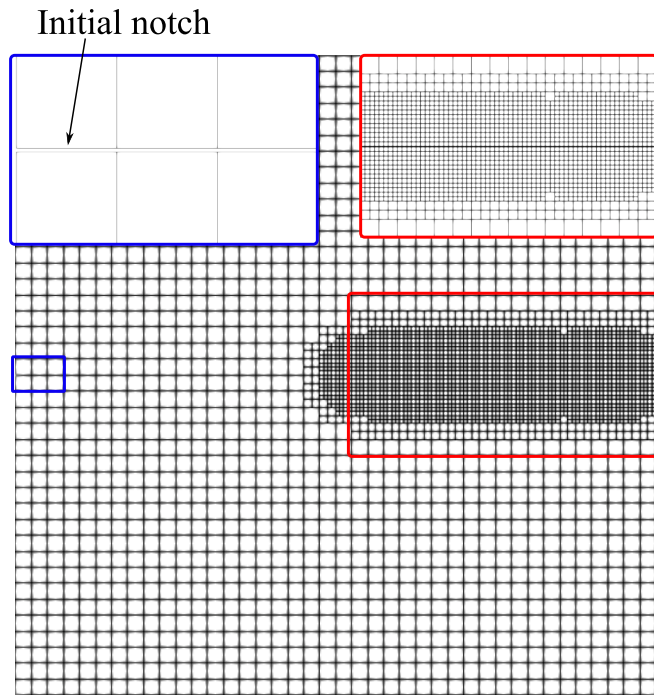


Fig. 6. Fixed mesh used in the single-edge notched tension test ($l_0/h = 2.4$).

2022) may give more objective adaptive solution, which however is not considered in the present study but will be explored in the future.

The calculation flow of the FVM-based phase field model with adaptive mesh refinement using the iterative staggered scheme is shown in Fig. 4 (for one loading step from t_i to t_{i+1}), where the residual tolerance is taken as $E_r = 10^{-7}$, and the maximum number of iterations is set to $n_{\max} = 1000$.

4. Numerical examples

To verify the correctness and effectiveness of the proposed algorithm, we present seven classical crack tests and compare the results with those available in the literature (mainly FEM simulations along with some experimental data). Particularly, the improvement of efficiency brought by the adaptive mesh refinement method is discussed. All 2D simulations presented in this section are performed in serial on an AMD® Ryzen® CPU R7-5800X @ 3.80 GHz with 32 GB RAM for fair comparisons with the Abaqus simulations. The parallel computing capability readily available in OpenFOAM (Cardiff et al., 2018) is only enforced in the 3D tests.

4.1. Single-edge notched tension test

The geometry and boundary conditions of the single-edge notched plate are shown in Fig. 5, where the initial crack is located at the left-center of the plate with a length of 0.5 mm and a width of 0.001 mm. The Young's modulus of the material is $E = 210$ GPa, the Poisson's ratio $\nu = 0.3$, and the critical energy release rate $G_c = 2.7$ N/mm. The phase field characteristic length is $l_0 = 0.015$ mm. These parameters are consistent with those used in Miehe et al. (2010a). The vertical displacement increment applied to the top surface is $\Delta u = 10^{-5}$ mm for the whole loading steps. Note that in this example, the AMR is not activated. A fixed mesh with local refinement is used instead as shown in Fig. 6 which contains 3864 quadrilateral cells with a minimum cell size of $h = 0.00625$ mm. The initial crack is modeled as a gap in the mesh domain (see Fig. 6) using the mesh-induced method (Krishnan et al., 2022).

The phase field contour during loading is shown in Fig. 7, which clearly shows a brittle fracture pattern: The crack starts to propagate from the tip at around $u = 5.54 \times 10^{-3}$ mm and quickly breaks the plate at $u = 5.92 \times 10^{-3}$ mm, forming a straight horizontal crack path. The reaction force–displacement curves are further shown in Fig. 8 with parametric studies on the influence of minimum mesh size h . In general, the present results agree well with the FEM results of Miehe et al. (2010a), both showing an initially elastic response up to a peak force of about 700 N followed by a drastic collapse.

For the mesh sensitivity analysis, we consider two extra cases with $h = 0.003125$ and 0.0015625 mm. It is seen that the peak force decreases with decreasing h , consistent with the observation in Molnr and Gravouil (2017), and exhibits a convergence trend. Additionally, it is worth mentioning that although the result with $h = 0.00625$ mm matches that of Miehe et al. (2010a) best, the latter employed a single-pass staggered scheme without convergence control. The use of smaller mesh sizes (i.e., $h = 0.003125$ and 0.0015625 mm) can result in a more accurate post-peak behavior, characterized by a straight drop in reaction force. Also implied in Fig. 8 is that the advantage of residual-controlled iterative staggered scheme is only effective when a certain minimum cell size is used.

To further compare the accuracy and efficiency of the proposed phase field model with existing FEM implementations, we use the code developed by Navidtehrani et al. (2021) using UMAT and HETVAL subroutines in Abaqus and simulate the same single-edge notched tension test. The influence of the ratio l_0/h on the peak reaction force is also studied. The following simulations adopt the same grid and residual-controlled staggered scheme for both FVM and FEM. Three cases of $l_0/h = 2.1$, 4.0, and 8.2, respectively, are analyzed. Figs. 9 and 10 present the fixed mesh and force–displacement curves, respectively, for the $l_0/h = 8.2$ case. The peak reaction forces and the time spent in the simulations are summarized and compared in Table 1.

Fig. 10 compares the force–displacement curves for the case $l_0/h = 8.2$ simulated by Abaqus and the present FVM model, together with that by Miehe et al. (2010a). It is observed that the present FVM result shows much better agreement with that of Miehe et al. (2010a), especially in the pre-peak stage and peak force. On the contrary, the Abaqus result

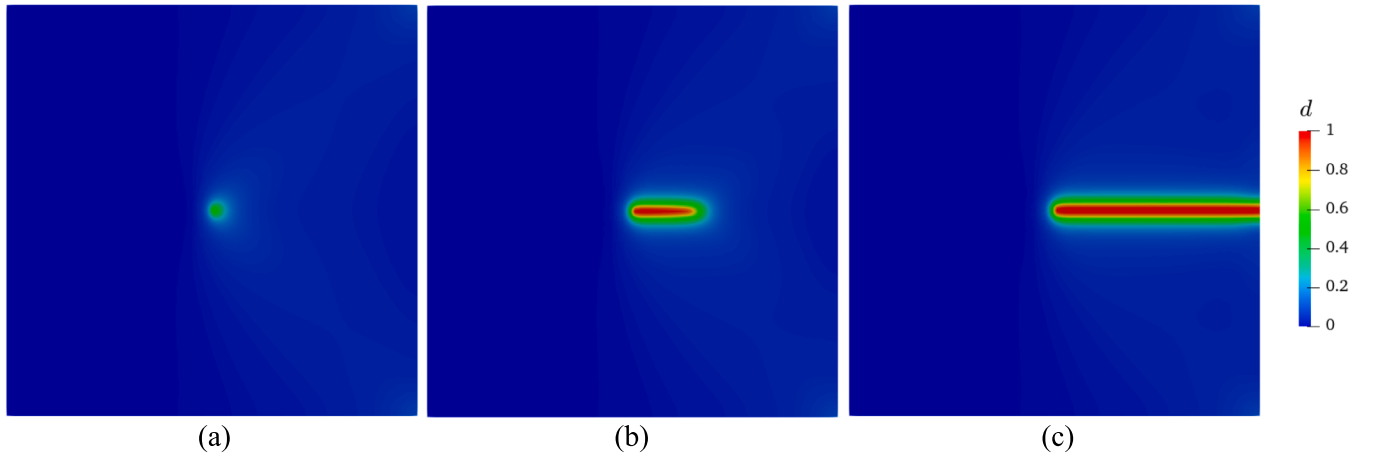


Fig. 7. Crack pattern of the single-edge notched tension test at displacements ($l_0/h = 2.4$): (a) $u = 5.54 \times 10^{-3}$ mm, (b) $u = 5.67 \times 10^{-3}$ mm, and (c) $u = 5.92 \times 10^{-3}$ mm.

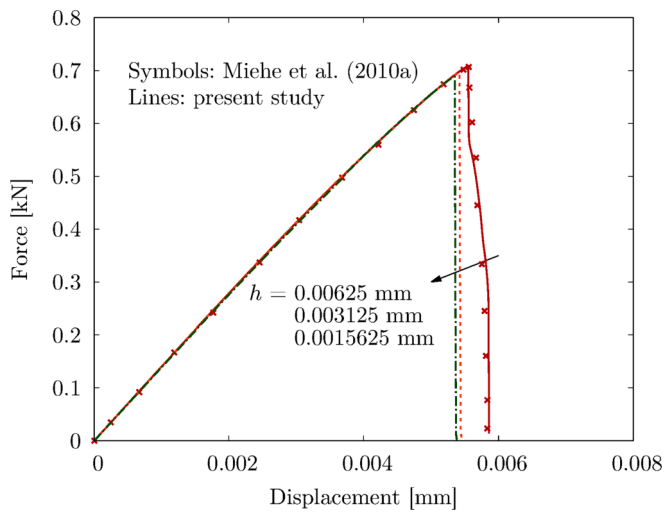


Fig. 8. Force-displacement curves of the single-edge notched tension test.

predicts a smaller elastic stiffness but a larger peak. The variations in post-peak behaviors are primarily attributed to the coarse discretization especially along the crack propagation path, as previously discussed. A similar observation was made by Mandal et al. (2019).

In addition, it can also be seen from Table 1 that for the same grid and staggered scheme, the FEM implementation is costlier than the present FVM model, especially when $l_0/h = 8.2$ (i.e., the finest mesh), the time cost in FVM is only 37 % of that in FEM, which highlights the advantage of using FVM for phase field modeling. Moreover, as the peak reaction force decreases with the decrease of minimum cell size or the increase of l_0/h , the FVM results show a better convergence performance than the FEM, which has also been confirmed by the study of Sargado et al. (2021).

4.2. Single-edge notched shear test

The geometry of this test is the same as the previous one, as shown in Fig. 5. The top boundary is displaced only in the horizontal direction at an increment of $\Delta u = 10^{-5}$ mm, and the vertical displacements of the lateral boundaries are prohibited as in Wu et al. (2020). Note that traction-free lateral boundaries are also used in some other studies (e.g., Ambati et al., 2015; Sargado et al., 2021) which will result in totally different post-peak responses. Likewise, a fixed mesh with initial local refinement as shown in Fig. 11 is used without AMR during loading. The

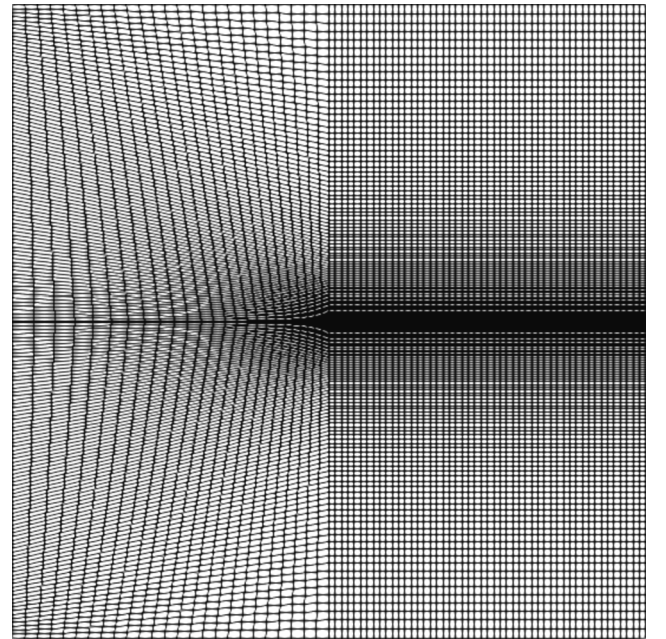


Fig. 9. Fixed mesh used in the single-edge notched tension test ($l_0/h = 8.2$).

parameters follow the previous test except that a different Poisson's ratio $\nu = 0.2$ is used in this shear test. To compare with the results in Wu et al. (2020), the spherical-deviatoric decomposition of strain energy (Amor et al., 2009) is used in this test.

The phase field evolution contour shown in Fig. 12 is consistent with that reported in Wu et al. (2020). In contrast to the straight crack path under tension, the crack path under shear develops from the initial crack tip at $u = 8 \times 10^{-3}$ mm but bends down to the lower right corner of the plate. According to the shear force–displacement curve shown in Fig. 13 (the case $l_0 = 0.015$ mm), the present model prediction highly agrees with the result in Wu et al. (2020), both in the pre- and post-peak stages. Compared with the curves under tension, the shear test results show less brittle responses.

We further investigate the effect of length scale on the simulation results. Based on the previous study (Bažant and Cedolin, 1983), another two tests with $l_0 = 0.03$ and 0.0075 mm are performed. To ensure convergent results, the ratio l_0/h of all the tests is controlled to be close and greater than 2. The results of the force–displacement curves are collectively shown in Fig. 13, which confirms that the peak reaction

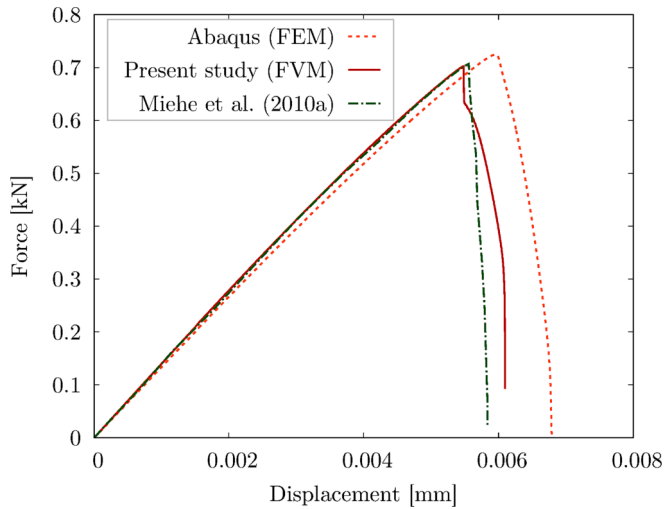


Fig. 10. Comparison of FEM and FVM solutions of the single-edge notched tension test ($l_0/h = 8.2$).

Table 1

Comparison of FEM and FVM for the single-edge notched tension test.

l_0/h	$h (\times 10^{-3} \text{ mm})$	Run time (s)		Peak reaction force (N)	
		FEM	FVM	FEM	FVM
2.1	7.200	858	530	759.15	714.12
4.0	3.740	1242	787	740.09	703.17
8.2	1.825	3067	1135	724.54	702.40

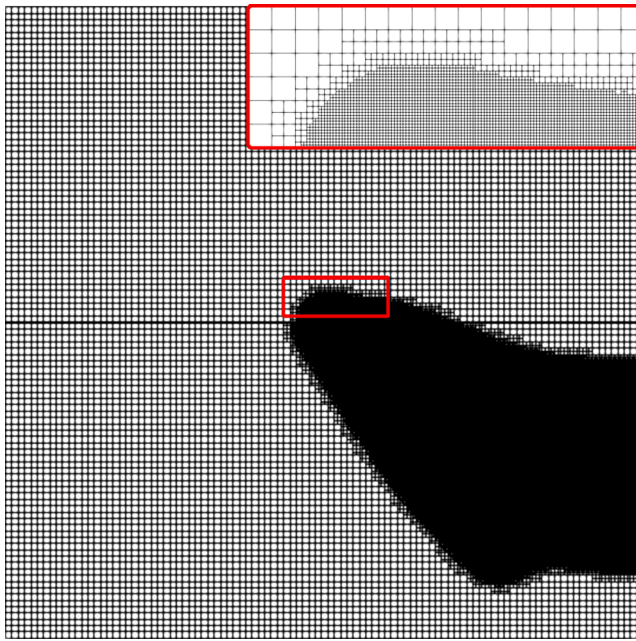


Fig. 11. Fixed mesh used in the single-edge notched shear test ($l_0 = 0.015 \text{ mm}$).

force decreases consistently as l_0 increases. The present results are generally consistent with the results given in Wu et al. (2020) and other similar studies.

4.3. Single-edge notched tension test with adaptive mesh refinement

To demonstrate the advantages of AMR, the single-edge notched

tension test is conducted with two fixed meshes and an adaptively refined mesh. The two fixed meshes correspond to the initial uniform mesh and the final adaptive mesh with local refinement, respectively. The parameters are basically the same as in section 4.1, except that $l_0 = 0.0075 \text{ mm}$. The meshes are shown in Fig. 14. The uniform mesh has 16,965 quadrilateral cells and a uniform cell size of 0.0077 mm. In AMR, the cell will be refined when its $d \geq 0.4$, and the refinement level is set to 2, which means that one cell would be split into 16 sub-cells.

The force–displacement curves are comparatively shown in Fig. 15. Clearly, the peak reaction force with AMR applied is smaller than that using the fixed uniform mesh, and the result with AMR matches notably better the result of Miehe et al. (2010a). A comparison between the result of AMR and that of the fixed locally refined mesh indicates that AMR maintains the computational accuracy. Compared with Fig. 10, the peak reaction force here is larger due to the use of a smaller l_0 . The post-peak behavior is not shown as a straight drop due to the use of insufficiently small cell sizes.

Finally, it is interesting to see from Table 2 that the final mesh in AMR has 22,308 cells, 31.5% more than the fixed uniform mesh. However, the simulation time with AMR is the least among the three cases. The use of AMR not only improves the result accuracy significantly but also saves computational time by reducing almost half of the time cost compared with the use of a fixed uniform mesh and 1/3 compared with the use of a fixed locally refined mesh.

4.4. L-shaped panel test

In this example, the crack propagation in a concrete L-shaped panel is simulated referring to the experimental test in (Winkler et al., 2001). The model geometry and boundary conditions are shown in Fig. 16(a). To approximate the point load in the experiment, a uniform displacement is applied to a small width (across 2 cells) in the numerical test. The Young's modulus $E = 25.85 \text{ GPa}$, Poisson's ratio $\nu = 0.18$, critical energy release rate $G_c = 0.089 \text{ N/mm}$, and length scale $l_0 = 1.1875 \text{ mm}$. The vertical displacement increment $\Delta u = 3.33 \times 10^{-4} \text{ mm}$ is applied until $u = 1 \text{ mm}$. The model is initially uniformly discretized with a cell size of 5 mm. The refinement level for AMR is 4. The final mesh is shown in Fig. 16(b), where the minimum cell size is 0.3125 mm.

Fig. 17(a)–(c) show the crack path evolution at different loading steps, roughly corresponding to the crack initiation, propagation, and final stages, respectively. The crack profile from the experimental tests (digitized from Ambati et al., 2015) is also superimposed in Fig. 17(c) for comparison. It can be concluded that the simulated crack path nicely lies in the range of the experimental results. Note that there is no predefined crack in this test. The results also prove that the present FVM-based phase field model can simulate arbitrary crack propagation problems.

4.5. Notched plate with a hole

This section simulates the crack initiation and propagation of a notched plate with a hole, following the experimental test presented by Ambati et al. (2015). The model geometry and boundary conditions are shown in Fig. 18(a), where the plate is fixed at the lower left pin, and loaded in the vertical direction through the upper left pin. The applied displacement increment is $\Delta u = 0.001 \text{ mm}$ until complete failure. The material of the plate is cement mortar, with the same parameters as those in the experiment (Ambati et al., 2015), i.e., the Poisson's ratio $\nu = 0.22$, Young's modulus $E = 4.9 \text{ GPa}$, and critical energy release rate $G_c = 2.28 \text{ N/mm}$. The length scale in the phase field simulation is $l_0 = 0.1 \text{ mm}$.

The initial mesh is generated using GMSH (Geuzaine and Remacle, 2009) with mixed triangle and quadrilateral cells, as shown in Fig. 18(b). The AMR is activated during loading with a refinement level of 3. The final mesh is given in Fig. 18(c) with a minimum cell size of 0.042 mm.

Fig. 19(a)–(c) present the crack paths at different displacements,

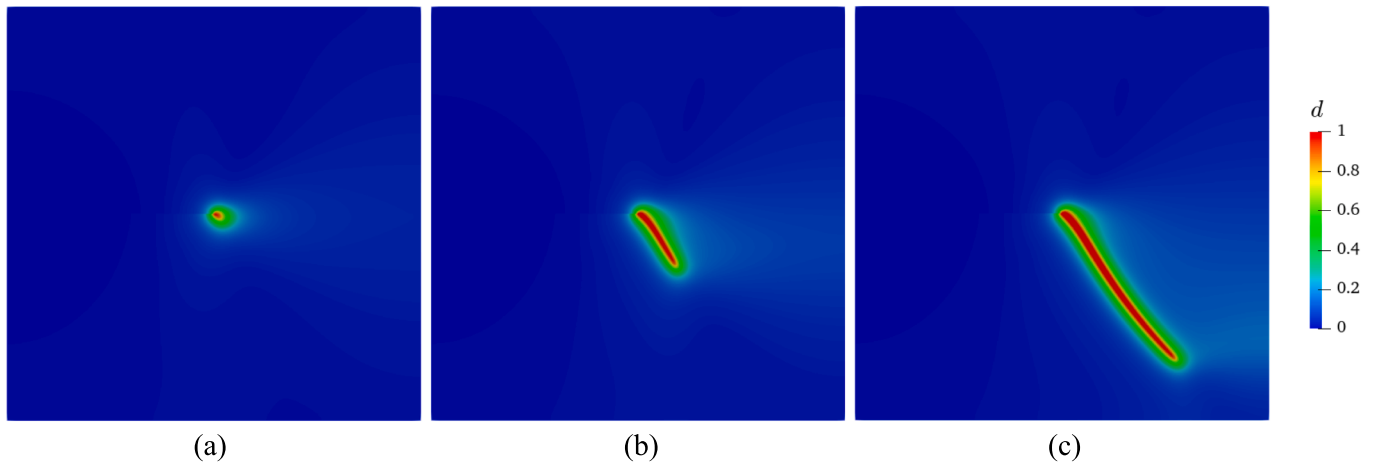


Fig. 12. Crack pattern of the single-edge notched shear test at displacements: (a) $u = 8 \times 10^{-3}$ mm, (b) $u = 10 \times 10^{-3}$ mm, and (c) $u = 11.8 \times 10^{-3}$ mm.

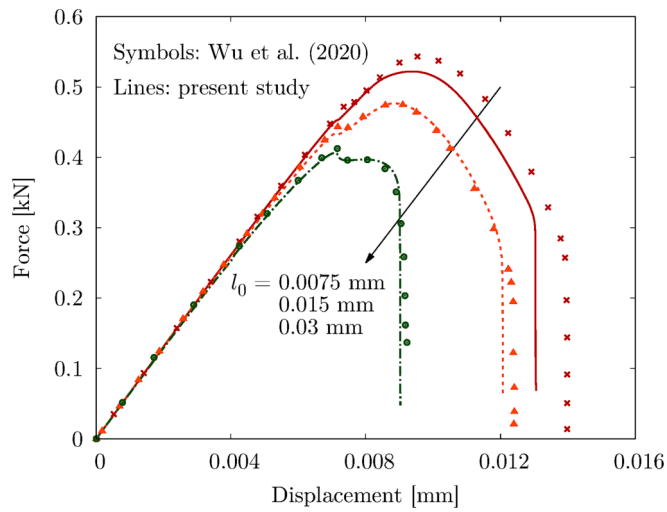


Fig. 13. Force-displacement curves of the single-edge notched shear test with different l_0 .

corresponding to the crack initiation, propagation, and complete failure stages, respectively. The final crack paths are consistent with those obtained in the experiment (Ambati et al., 2015), as shown in Fig. 19(d).

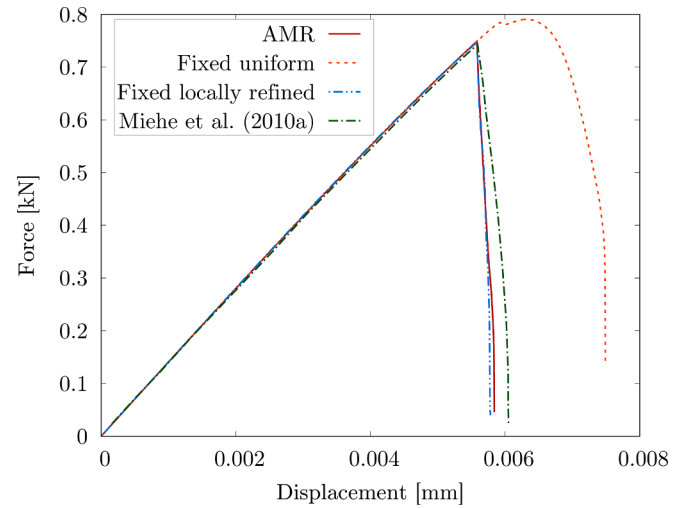


Fig. 15. Force-displacement curves of the single-edge notched tension test with or without AMR.

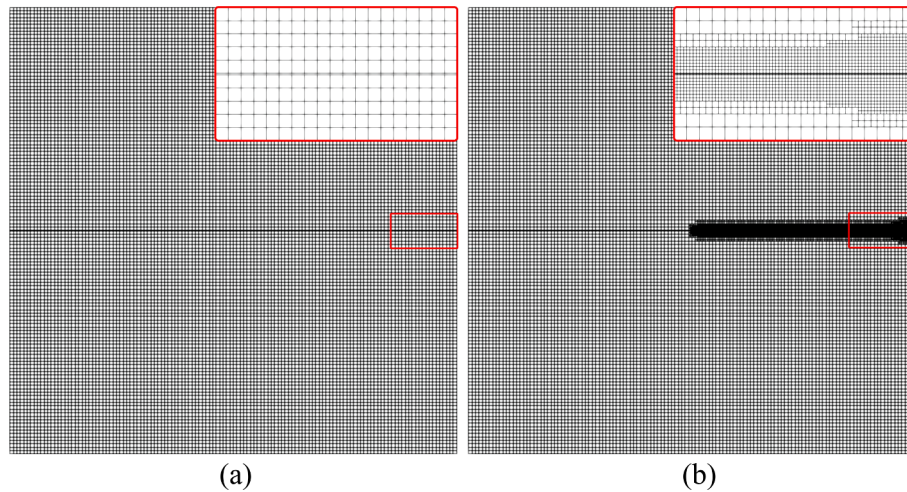


Fig. 14. Meshes used in the single-edge notched tension test: (a) fixed uniform mesh, and (b) final mesh after AMR (same as the fixed locally refined mesh).

Table 2
Summary of the single-edge notched tension test with or without AMR.

Mesh type	Initial / final cell numbers	Run time (s)	Speedup
AMR	16,965 / 22,308	3862	1.97
Fixed uniform mesh	16,965 / 16,965	7599	1
Fixed locally refined mesh	22,308 / 22,308	5790	1.50

4.6. Three-dimensional single-edge notched tension and shear tests

The last two examples demonstrate that the present FVM-based phase field model can be easily extended to 3D simulations. The single-edge notched tension and shear tests are performed for this purpose. For the tension test, the 3D model has a thickness of 0.1 mm, and the other parameters are the same as in the 2D test presented in section 4.1 ($l_0 = 0.015$ mm).

The model is initially uniformly discretized with the cell size $h = 0.025$ mm (the out-of-plane cell size is 0.02 mm), and the AMR is evoked during loading to accelerate the computation with the refinement level of 2. Fig. 20 shows the meshes at different time steps. It is seen that the mesh refinement starts from around the crack tip at $u = 6.016 \times 10^{-3}$ mm and shortly develops to form a band around the full crack plane at $u = 6.205 \times 10^{-3}$ mm. The difference in mesh density afterward is insignificant. The evolution of phase field iso-volume ($d > 0.9$) in Fig. 21 confirms the path of crack propagation, similar to the results in Xu et al. (2022) and Dinachandra and Alankar (2022). The force–displacement curve after normalization by thickness is found identical to that in the 2D test (i.e., Fig. 8), which is therefore not repeated here.

For the 3D shear test, the model has a thickness of 0.05 mm and the other parameters are the same as in the 2D test presented in section 4.2 ($l_0 = 0.015$ mm). The model is initially uniformly discretized with the cell size $h = 0.01$ mm (the out-of-plane cell size is 0.025 mm), and the

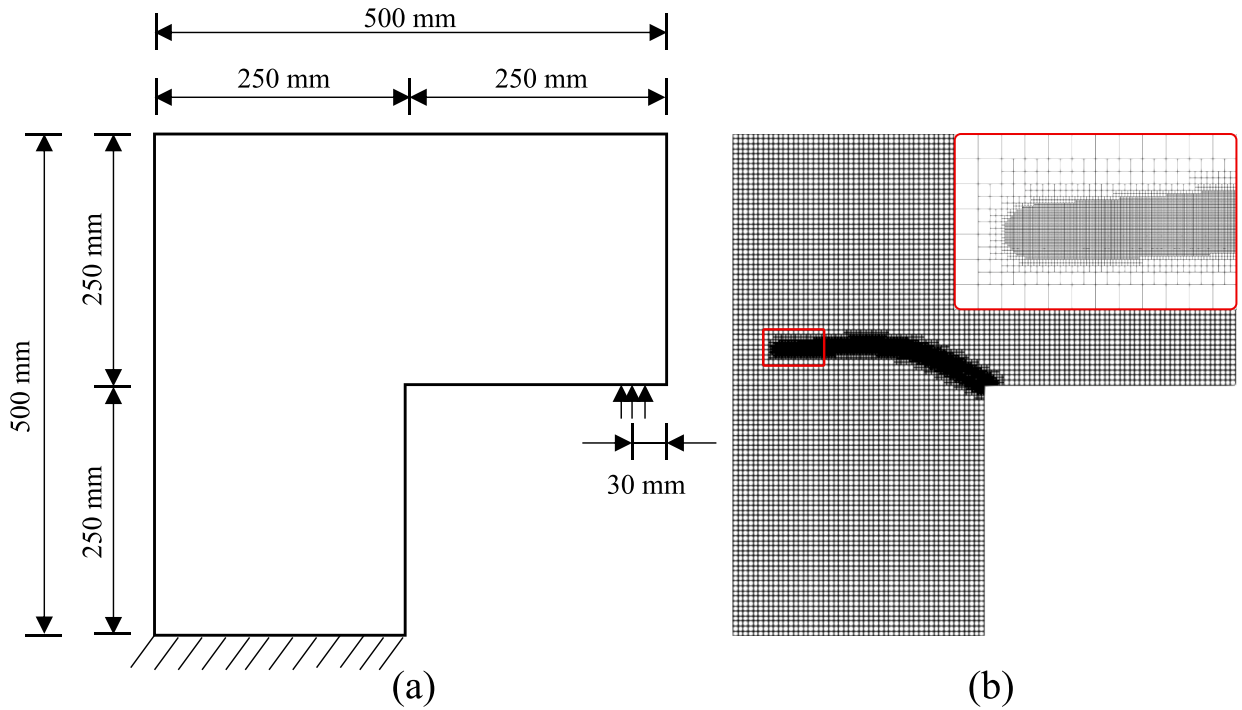


Fig. 16. L-shaped panel test: (a) geometry and boundary conditions, and (b) final mesh.

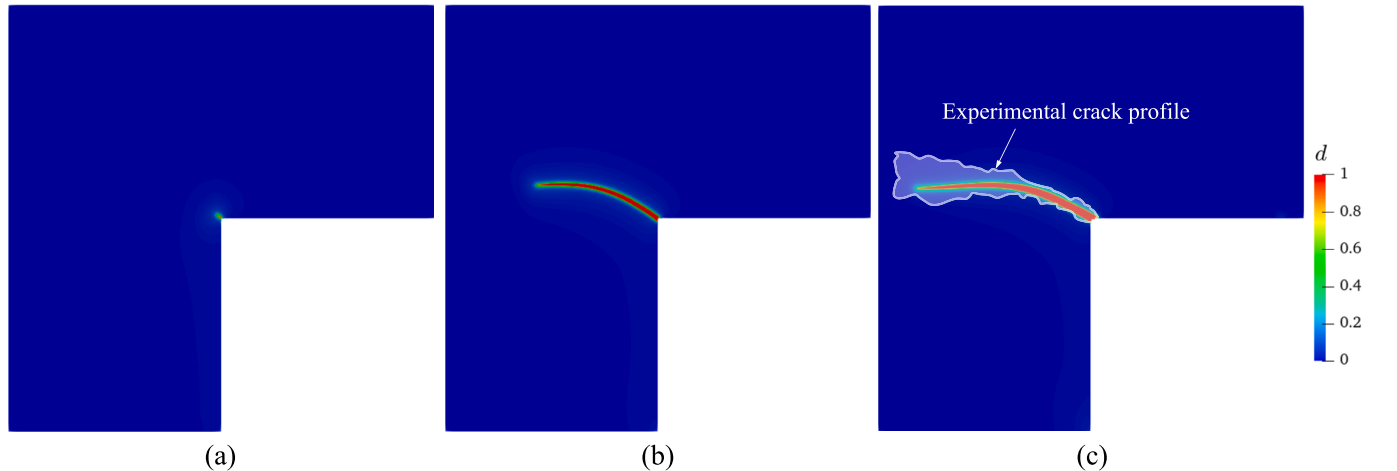


Fig. 17. Crack pattern of the L-shaped panel test at displacements: (a) $u = 0.25$ mm, (b) $u = 0.55$ mm, and (c) $u = 1$ mm. The experimental crack profile is superimposed in (c).

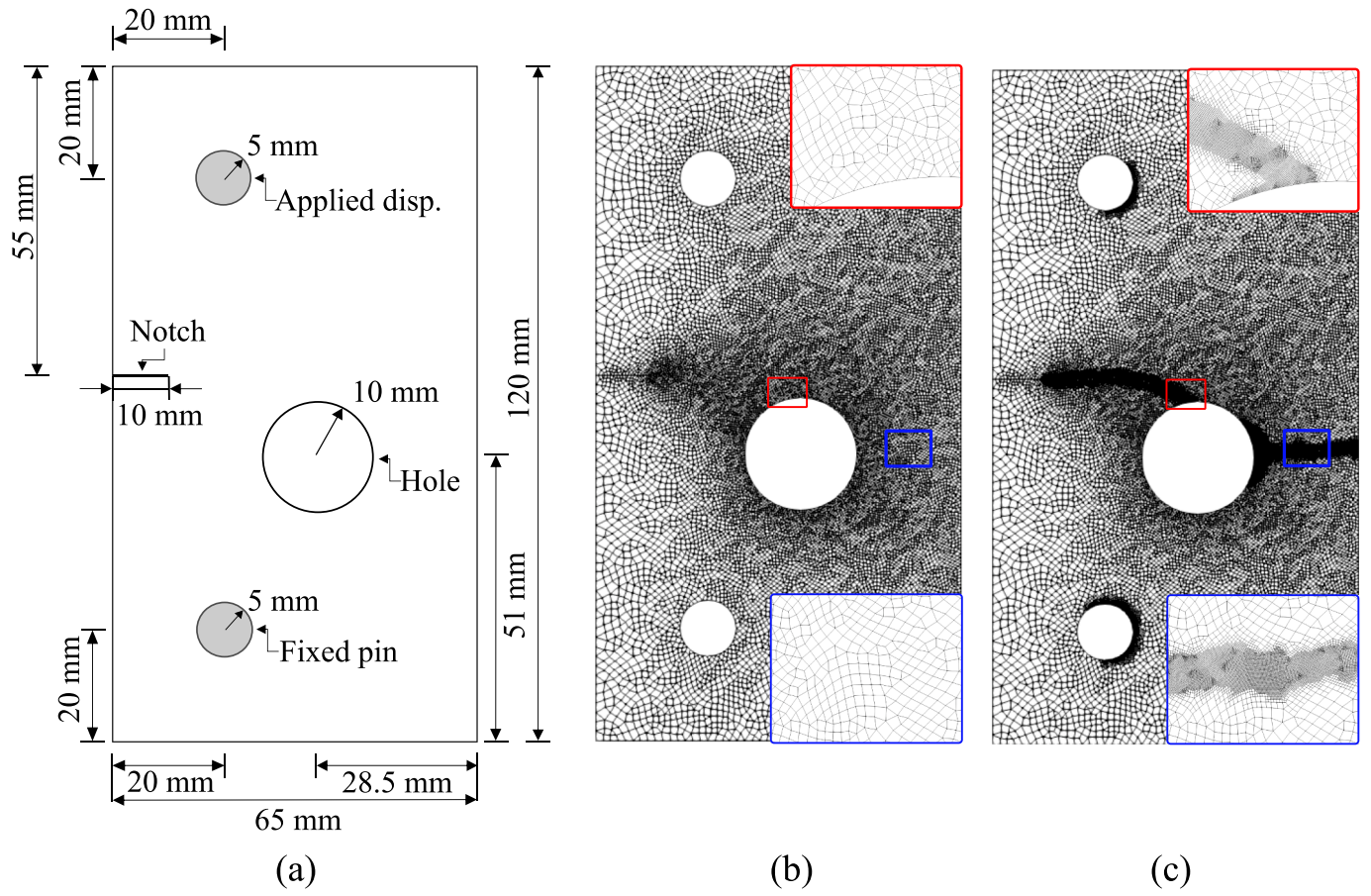


Fig. 18. Notched plate with a hole: (a) geometry and boundary conditions, (b) initial mesh, and (c) final mesh.

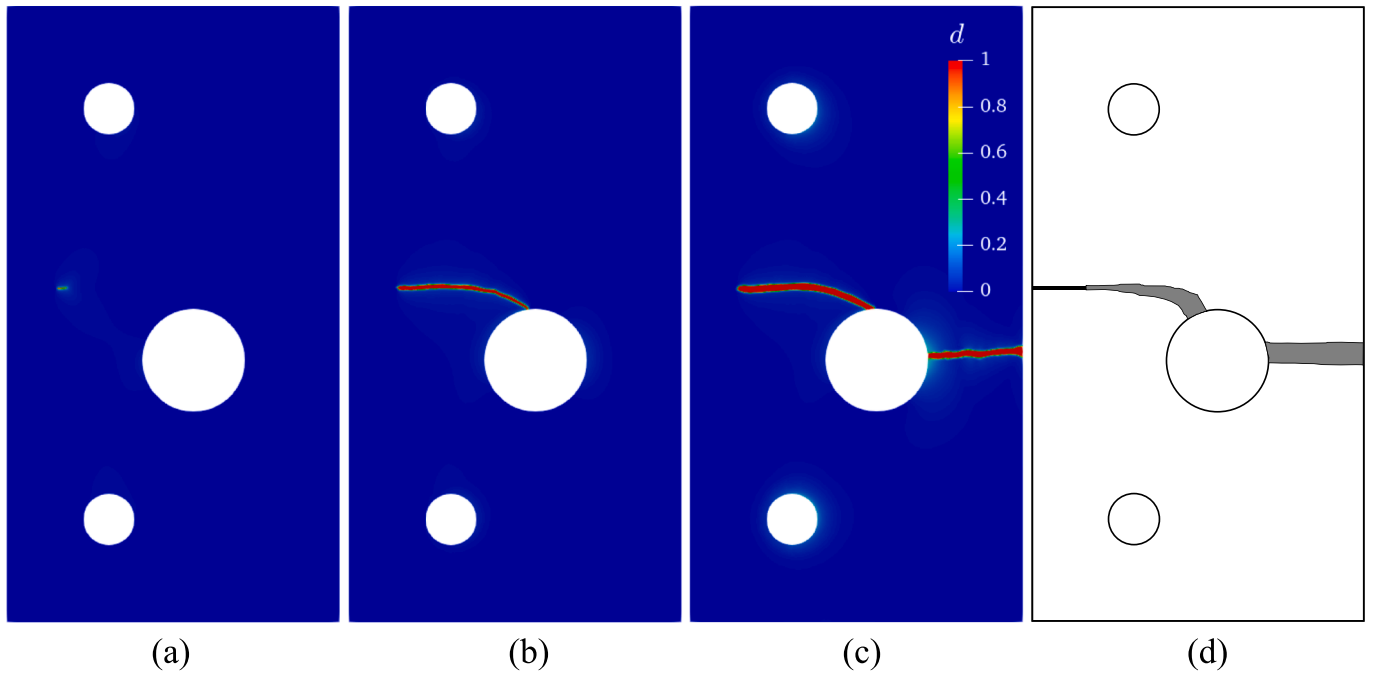


Fig. 19. Crack pattern of the notched plate with a hole at displacements: (a) $u = 0.4$ mm, (b) $u = 1.085$ mm, (c) $u = 2.145$ mm, and (d) experimental crack profile (adapted from Ambati et al., 2015).

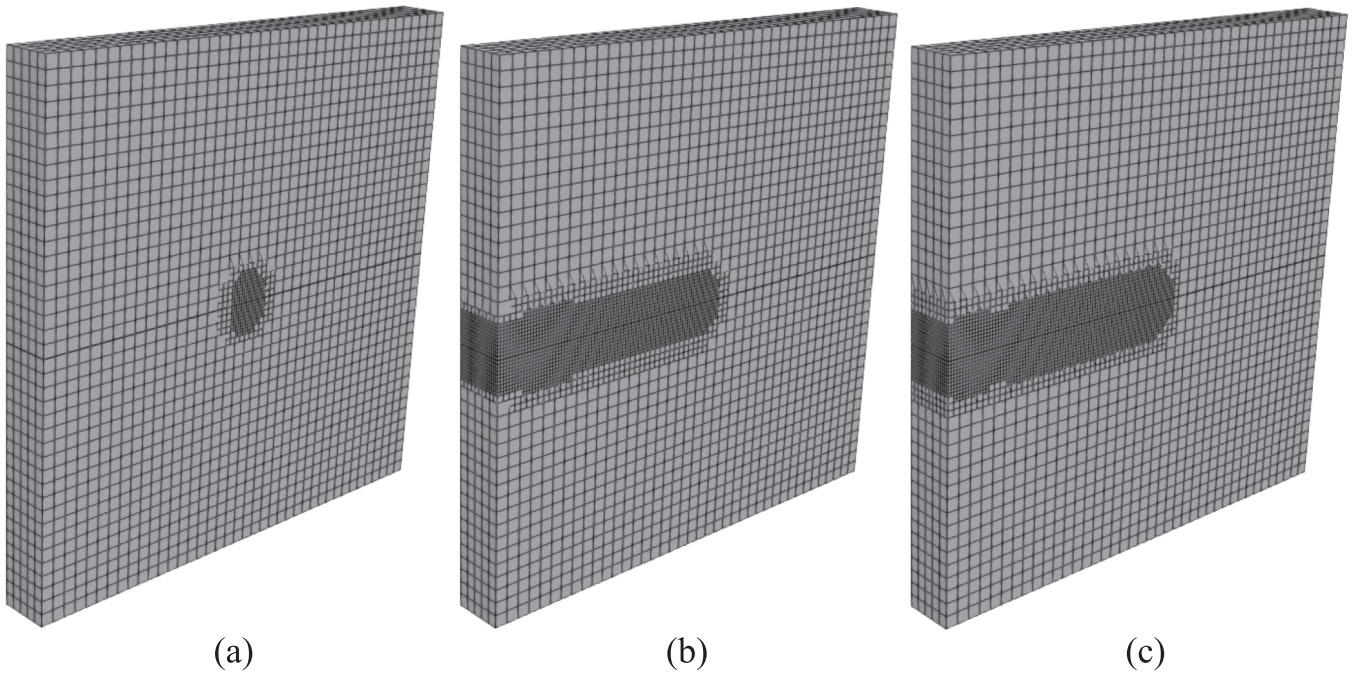


Fig. 20. Meshes in the 3D single-edge notched tension test at displacements: (a) $u = 6.016 \times 10^{-3}$ mm, (b) $u = 6.205 \times 10^{-3}$ mm, and (c) $u = 9.0 \times 10^{-3}$ mm.

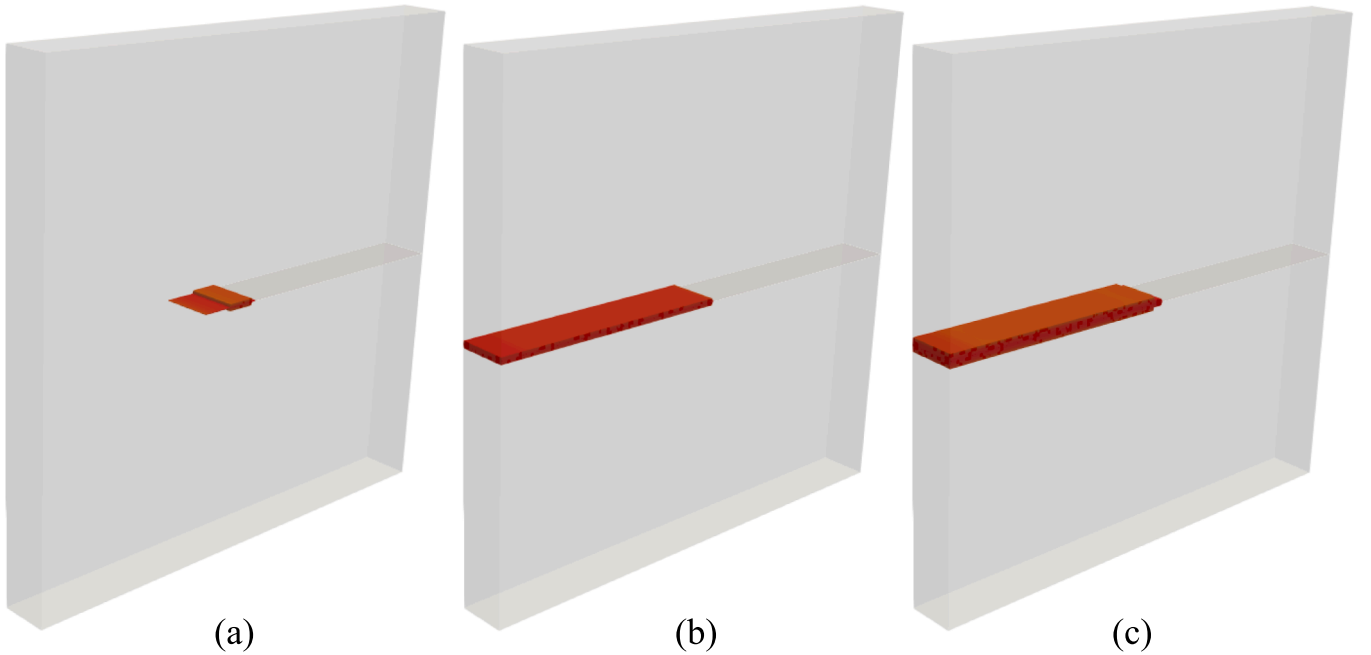


Fig. 21. Phase field *iso-volume* ($d > 0.9$) in the 3D single-edge notched tension test at displacements: (a) $u = 6.016 \times 10^{-3}$ mm, (b) $u = 6.205 \times 10^{-3}$ mm, and (c) $u = 9.0 \times 10^{-3}$ mm.

AMR is evoked during loading to accelerate the computation with the maximum refinement level set to 3. Fig. 22 shows the meshes at different time steps. It is seen that the mesh refinement starts from around the crack tip at $u = 8.0 \times 10^{-3}$ mm and gradually evolves into a curved band around the crack surface at $u = 12.1 \times 10^{-3}$ mm. The evolution of phase field *iso-volume* ($d > 0.9$) in Fig. 23 confirms the path of crack propagation. By examining the morphology of the local meshes around the crack tip, it can be observed that in the 3D AMR, a quadtree refinement scheme is activated when a structural hexahedral mesh is used. The force–displacement curve after normalization by the thickness is found quite close to that in the 2D test (i.e., Fig. 13), which is therefore not

repeated here.

5. Conclusions

A cell-centered FVM implementation of the phase-field model with AMR for brittle fracture using the iterative staggered scheme is presented in the study. Two commonly used strain energy density decomposition methods, namely, the spherical-deviatoric decomposition and spectral decomposition, are considered. The proposed model has been applied to simulating several classical crack propagation problems, including the single-edge notched tension and shear tests, L-shaped

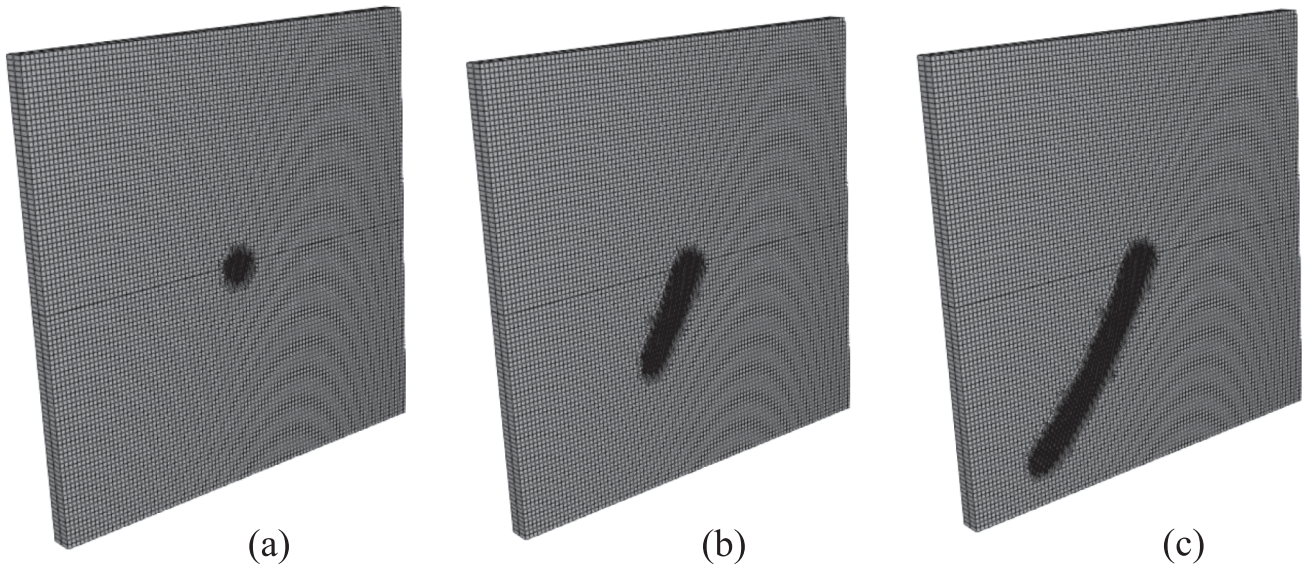


Fig. 22. Meshes in the 3D single-edge notched shear test at displacements: (a) $u = 8.0 \times 10^{-3}$ mm, (b) $u = 10.7 \times 10^{-3}$ mm, and (c) $u = 12.1 \times 10^{-3}$ mm.

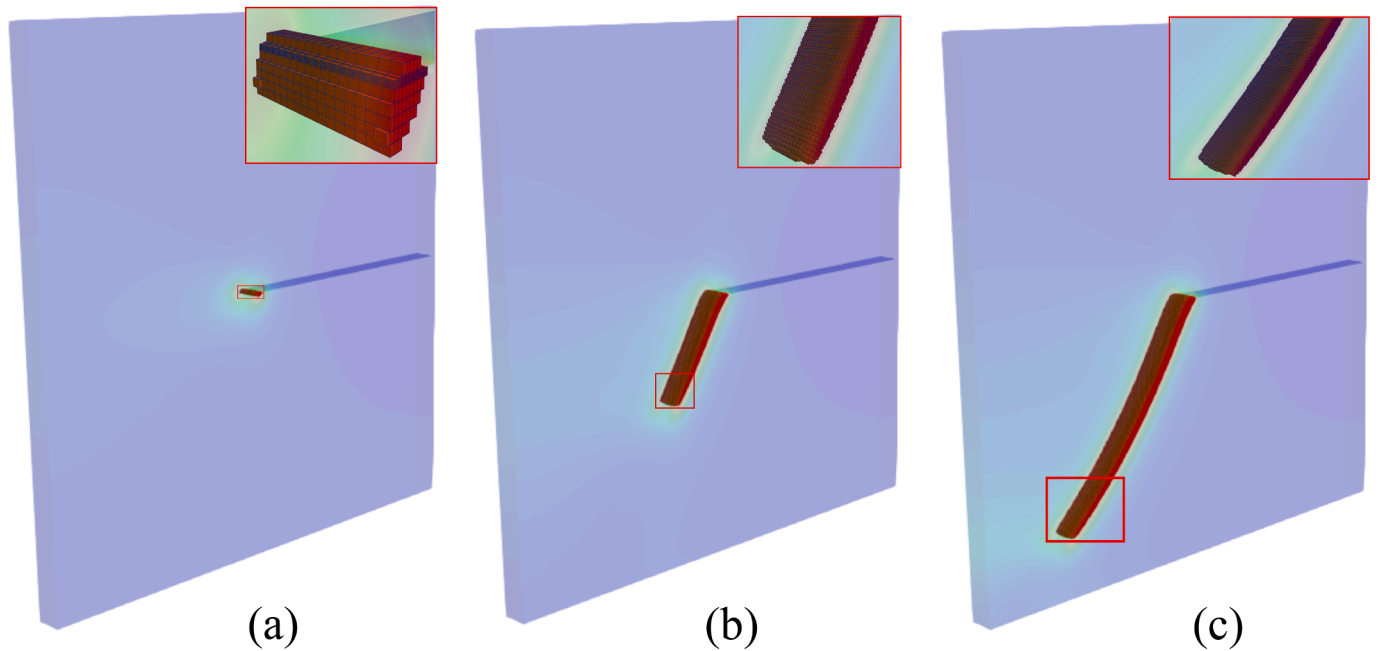


Fig. 23. Phase field *iso*-volume ($d > 0.9$) in the 3D single-edge notched shear test at displacements: (a) $u = 8.0 \times 10^{-3}$ mm, (b) $u = 10.7 \times 10^{-3}$ mm, and (c) $u = 12.1 \times 10^{-3}$ mm.

panel test, and test on a notched plate with a hole. By comparing with the results from available experiments and FEM studies, it is concluded that the FVM-based phase-field model is a promising tool to solve these crack propagation problems and could be advantageous to the alternative FEM-based model in terms of efficiency in some case studies. The isotropic AMR algorithm adopted in the study can effectively improve the computational accuracy meantime significantly reduce the computational cost.

Another feature of FVM is that it has been extensively applied in solving continuity or transport equations, which means that the FVM-based phase-field model could be conveniently coupled with fluid flow to solve multiphysics problems such as hydraulic fracturing. This will be pursued in a future study.

CRediT authorship contribution statement

X.L. Yang: Conceptualization, Software, Data curation, Investigation, Writing – original draft. **N. Guo:** Supervision, Conceptualization, Methodology, Funding acquisition, Investigation, Writing – review & editing. **Z.X. Yang:** Conceptualization, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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